RESEARCH REPORT · VITKI

Fully tested finding

# Monophthong spellings become rarer on the runestones – from 69 to 22 percent between about 1000 and 1115

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**Timeline of the study:** The pattern was found and the checks of the measurement were added 11 October 2026 · The material was frozen 29 October 2026 · The coding was checked 31 October 2026

## ABSTRACT

**Background and aim.** The East Norse monophthongisation, in which the diphthong æi became ē (*stæinn* > *stēn*), has been dated from the spelling of the runestones: *stin* instead of *stain* has been read as a sign that the sound change had reached the district. The monophthong spelling should then become more common in the Mälaren region during the eleventh century. The question is how the share changes over time with place and carver held constant.
**Material.** 649 dated Swedish Viking-Age runestones with at least one word that has æi in the Scandinavian Runic-text Database (Rundata), 1,278 such words in all.
**Method.** Logistic regression with the style group's midpoint as time, the stone's location as a spatial smooth and the carver as a random effect; Holm correction over 14 traits; replication in two halves; eight sensitivity analyses; twelve checks of the measurement and one of the coding; adjustment for all 48 associations the project reports.
**Results.** The share of stones with a monophthong spelling falls from 69 percent around 998 to 22 percent around 1115, 48 percentage points (95 % interval 22–66), p = 0.005, Holm-corrected p = 0.028. With the years drawn within the style groups' intervals the decline is 22–33 percentage points. Same direction and significant in both halves; every sensitivity analysis holds; adjusted for all associations q < 0.001. The decline is seen word by word and in both Uppland and Södermanland, but it lies at the beginning and end of the period: without the earliest and the latest style group the trend is not significant, and in Uppland the two ends are largely two carvers, Gunnar and Öpir.
**Conclusion.** The spelling moved in the opposite direction to the sound change it has been used to date. The late Upplandic spelling *stain* probably reflects a written norm rather than the pronunciation, and the runestone spelling in the Mälaren region therefore cannot by itself show when the monophthongisation reached it.

**Keywords:** *runestones, Viking Age, Runic Swedish, monophthongisation, spelling, written norm, carvers, stylistic chronology, logistic regression*

HIGHLIGHTS

- The runestones wrote the monophthong less and less often, although the language was heading the other way.
- The late Upplandic spelling *stain* looks like a written norm, carried by carvers such as Öpir, and not like the pronunciation.
- The spelling on the stones of the Mälaren region therefore cannot by itself show when the monophthongisation arrived there.

## 1. Introduction

There is agreement that the East Norse monophthongisation, in which the diphthongs æi, au and øy became long monophthongs, began in Denmark in the tenth century, and the evidence is the spelling of the runic inscriptions: the digraphs are replaced by a single rune, so that *stain* becomes *stin* (Marklund 2018, p. 12, with references). From the spelling of the runestones Elias Wessén assumed that the sound change reached Östergötland and Västergötland around the year 1000 and central Sweden towards the end of the eleventh century (Wessén 1968, pp. 32 f., after Marklund 2018, p. 12). Magnus Källström (2014) has read the spelling *sten*, with a dotted i-rune, as a sign that æi had become [e:] in Västmanland, around Uppsala, in western Uppland and at two places in Södermanland, and he frames the question as one of written norm and dialect (after Marklund 2018, p. 13). That the carvers spelled differently, and that spelling can tell them apart, has long been known (Thompson 1975).

We have found no study that measures how the share of monophthong spellings changes over time with place and carver held constant. If the spelling followed the pronunciation, the share should rise in the Mälaren region during the eleventh century. The aim is to measure the change and to test whether it holds.

## 2. Pre-registration: what was fixed in advance and what is exploratory

No analysis plan was locked before the results were known; the finding was discovered in a systematic survey of 14 traits, with the same model, measure of time, correction and halves for all, fixed in the program code before the run. The trait was added on 11 October 2026. On the same day, after the pattern had been seen, twelve checks of the measurement were added against conceivable objections; all are reported in 5.4, including those that do not hold. The p-value is corrected for the 14 traits and for all 48 associations, and the halves were fixed independently of the patterns. The

check of the coding in 5.4 was made on 31 October 2026, while the report was being written. Section 5.5 is exploratory, computed after the pattern was known, with unadjusted p-values.

## 3. Material

- **Source.** The Scandinavian Runic-text Database, Department of Scandinavian Languages, Uppsala University: Rundata (2014 edition, updated 2018), retrieved 2026-10-04; derived corpus computed 2026-10-29.
- **The stones.** Viking-Age runestones (no other objects) in the Swedish provinces; Skåne and Blekinge are counted as Denmark in Rundata.
- **The words.** Words with æi in Rundata's normalisation (*ræisa*, *stæinn*, *þæiR*, *bæði* and others), compared word by word with the transliteration when the two texts have the same number of words; proper names and uncertain readings are not counted. A word is spelled with a *diphthong* when the transliteration has ai, ia, ei or ae, otherwise with a *monophthong*. *stæinn* makes up 48 percent of the 1,278 words.
- **The trait.** A stone has a *monophthong spelling* when at least one of its æi words is spelled with a monophthong. The trait can be determined on 1,091 stones; 649 of them are dated, and 363 have a monophthong spelling.
- **Time.** The style group's midpoint after Gräslund (2006): RAK 998, Fp 1030, Pr1 1030, Pr2 1035, Pr3 1065, Pr4 1085, Pr5 1115. The style groups rest on the ornament, not on the spelling.
- **The carver.** The carver Rundata gives without a question mark; stones without a secure carver are grouped by parish.

## 4. Method

**4.1 The model.** Logistic regression (mgcv, R 4.6.1; Wood 2017) with the trait against the style-group midpoint as a smooth curve, the stone's location as a spatial smooth and the carver as a random effect, so that a prolific carver's habit is not counted as a change. The curve is shown as the adjusted share for a typical carver, with 95 % intervals from 300 draws of the coefficients. The uncertainty of the dating is tested by drawing each stone's year evenly within its style group's interval, ten times.

**4.2 Correction and replication.** Holm correction (Holm 1979) over the 14 traits. The material was divided once, for all the project's analyses, into two halves parish by parish, and the change must hold in each half.

**4.3 Sensitivity analyses.** The same fixed series as for every association, with time in steps of 50 years: the raw model; adjusted for province; the carver as a random effect; the parish as a random effect; the stone's location as a spatial surface (only stones that stand in place); each half on its own; and without one province at a time (provinces with at least 30 stones). An analysis holds when p is below 0.05 and the direction is the same as in the adjusted model; without a province, significance is required only where at least two thirds of the stones remain. The p-values are ad-

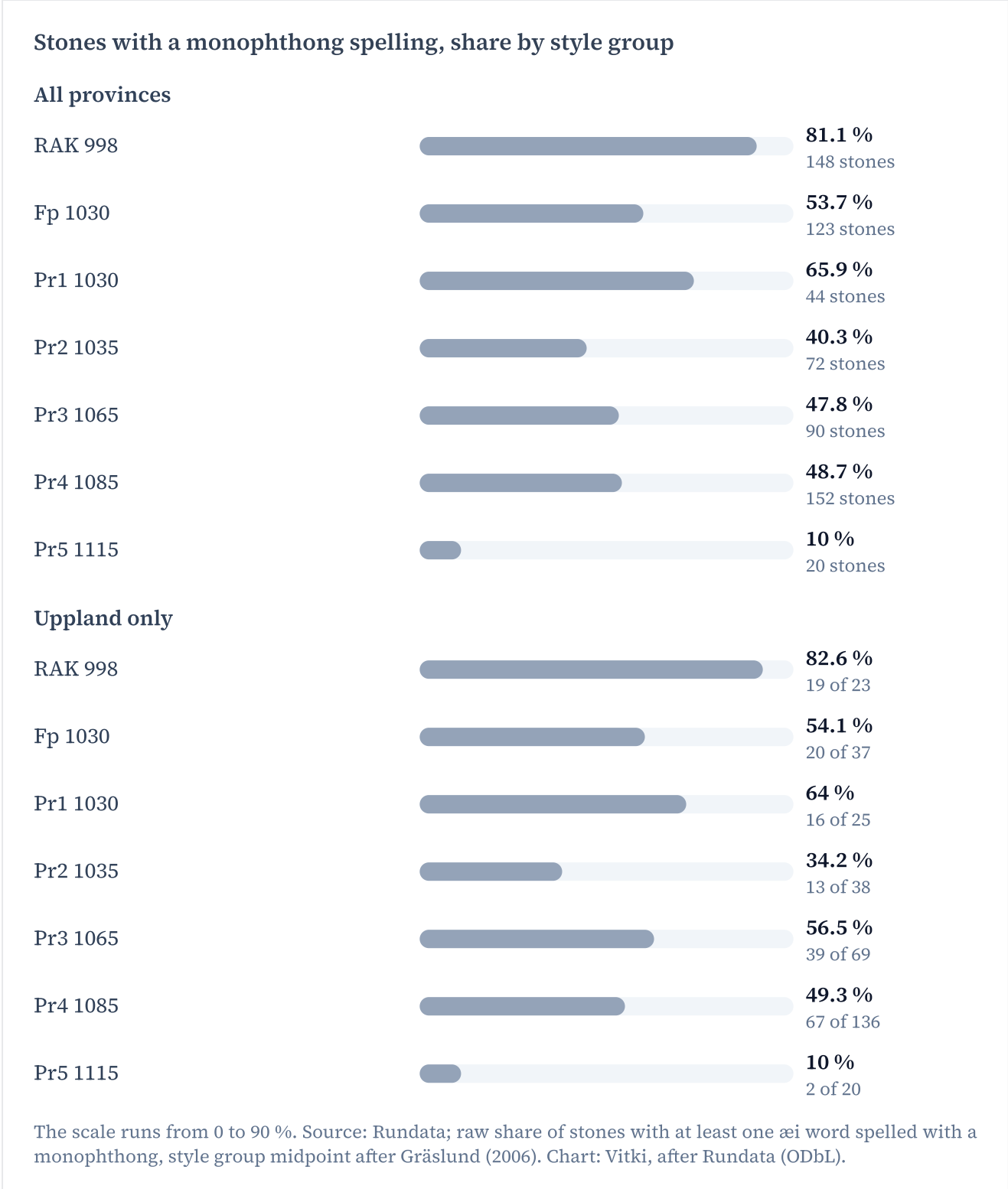
justed together for all 48 associations (Benjamini & Hochberg 1995); the adjusted value is called q. The assumptions are checked with simulated residuals (DHARMa; Hartig 2022).

**4.4 Checks of the measurement.** Twelve variants of the same straight trend (5.4) and a check of the coding with a stricter rule.

**4.5 Deviations and errors.** No plan existed, so no deviations can be reported. While the report was being written, a random sample of the words showed that the rule also counts words spelled with a alone or without a vowel as monophthongs; the result was therefore recomputed with a stricter coding (5.4) and does not change.

## 5. Results

**5.1 The main test.** With location and carver constant the share falls from 69 percent around 998 to 22 percent around 1115, 48 percentage points (95 % interval 22–66), p = 0.005, Holm-corrected p = 0.028. The curve does not fall evenly: from 69 to 58 percent up to about 1030, hardly moving until about 1070 (55 percent), then down again. Half A: 343 stones, from 70 to 24 percent, p = 0.049. Half B: 306 stones, from 69 to 33 percent, p = 0.015. In plain words: around 1000 about seven stones in ten had at least one word spelled with a monophthong, around 1115 about two in ten. The sensitivity analyses' odds ratio of about 0.5 per 50 years means that the odds of a monophthong spelling roughly halve in fifty years.




**5.2 The uncertainty of the dating.** When each stone's year is drawn within its style group's interval, the change is significant in 9 of 10 refits and has the same shape in all of them, but it becomes smaller: 22–33 percentage points.

**5.3 The sensitivity analyses** (649 stones, 363 with a monophthong spelling; odds ratio per 50 years). All 8 hold:

- Raw model: 0.44 (0.35–0.56), $p < 0.001$.

- Adjusted for province: 0.55 (0.39–0.76), p < 0.001. That corresponds to 68 percent in 1000 against 42 percent in 1100, a difference of about 25 percentage points (probably 12–39).
- Carver as a random effect (315 carvers and parish groups): 0.52 (0.33–0.81), p = 0.004. Parish as a random effect (326 parishes): 0.49 (0.34–0.70), p < 0.001.
- Location as a spatial surface (390 stones that stand in place): 0.54 (0.34–0.88), p = 0.014.
- Half A (340 stones): 0.60, p = 0.025. Half B (309 stones): 0.49, p = 0.005. The halves are divided by the same rule but computed by another part of the program, in which parishes with the same mean dating come in a slightly different order, so 9 stones fall in another half than in 5.1 (340 and 309 instead of 343 and 306).
- Without one province at a time (4 provinces): the direction holds without each of them; weakest without Uppland, when 301 stones remain, fewer than two thirds: 0.53 (0.24–1.17), p = 0.12.
- The assumptions: residual distribution p = 0.25, overdispersion p = 0.26, spatial pattern p = 0.26; no deviations.
- Adjusted for all 48 associations: q < 0.001, one of 15 that hold in every sensitivity analysis.

**5.4 The checks of the measurement.** The direction is the same in all twelve; nine are significant and three are not (odds ratio per 50 years):

- Per stone: odds ratio per 50 years 0.55 (0.37–0.83), p = 0.005. Per word: odds ratio per 50 years 0.54 (0.38–0.78), p < 0.001.
- *stæinn* only: odds ratio per 50 years 0.59 (0.38–0.92), p = 0.021 (569 stones). Other æi words only: odds ratio per 50 years 0.64 (0.41–0.99), p = 0.044 (531 stones).
- Only stones with a secure carver: odds ratio per 50 years 0.46 (0.24–0.87), p = 0.018 (270 stones); only stones without: odds ratio per 50 years 0.55 (0.35–0.87), p = 0.010 (379 stones); without the carver in the model: odds ratio per 50 years 0.54 (0.43–0.68), p < 0.001.
- Uppland only: odds ratio per 50 years 0.42 (0.26–0.67), p < 0.001 (348 stones). Södermanland only: odds ratio per 50 years 0.31 (0.12–0.84), p = 0.020 (113 stones).
- **Not significant:** only signed stones, odds ratio per 50 years 0.91 (0.39–2.15), p = 0.83 (106 stones); within the same carver, odds ratio per 50 years 0.50 (0.23–1.10), p = 0.084 (11 carvers with stones in at least two style groups, 174 stones); without the earliest and the latest style group, odds ratio per 50 years 0.67 (0.42–1.08), p = 0.098 (481 stones).
- **The coding.** Of the 599 words the rule counts as monophthongs, 79 are spelled with a alone or without a vowel (most often *stan*, *rasa* and *þaR*). With a stricter rule, in which only i or e in the place of the diphthong counts as a monophthong and the other words as uncertain, 28 stones change group and 16 drop out. The odds ratio becomes 0.53 adjusted for province (against 0.54 with the rule), 0.57 and 0.46 in the halves and 0.58 in Uppland, all with p below 0.01.

**5.5 The ends and the carvers (exploratory, descriptive).**

- In Uppland 19 of 23 RAK stones have a monophthong spelling, and 16 of them were carved by Gunnar, who writes a monophthong on 19 of his 20 stones. Of the 20 Pr5 stones, 2 have a monophthong spelling, and 16 are Öpir's. In between, the share lies between 34 and 64 percent in every style group.

- Öpir writes a monophthong on 11 of 21 stones in Pr4 but on 1 of 16 in Pr5 (Fisher's exact test, p = 0.004, unadjusted), the clearest change within one carver in the material.
- It is the i-spelling (*stin*) that disappears, from 63.1 percent of the æi words on the RAK stones to 3.3 percent on the Pr5 stones. The e-spelling with a dotted i-rune (*sten*) is rare in every style group, 2.0–9.9 percent, most common on the Pr4 stones.

## 6. Discussion

**What the finding says.** On the runestones the monophthong spelling becomes rarer over time, with location and carver constant, in both halves and in Uppland and Södermanland. If the spelling followed the pronunciation, the Mälaren region would have had the monophthong early and lost it, and nobody assumes that: Old Swedish has the monophthong. The most reasonable reading is that the spelling did not follow the pronunciation. The early carvers often wrote a single rune for the diphthong, the late Upplandic ones almost always *ai*. That looks like a written norm that took hold, carried by carvers such as Öpir, rather than a change in speech.

**Against earlier research.** Wessén dated the sound change from the spelling of the runestones, but in the Mälaren region this spelling moves in the wrong direction to bear such a dating. It supports Källström's (2014) way of keeping written norm and dialect apart, and it fits with the fact that the i-spelling disappears while the e-spelling, which Källström reads as a monophthong, remains. It does not say when the monophthongisation reached the Mälaren region, only that the runestones' *stin* and *stain* cannot show that on their own.

**Weaknesses, in order of weight.**

1. **The ends carry the decline.** Between about 1030 and 1070 the share hardly moves, and without the earliest and the latest style group the trend is not significant (p = 0.098). Pr5 has only 20 stones.
2. **In Uppland the ends are two carvers.** Gunnar accounts for 16 of 23 RAK stones and Öpir for 16 of 20 Pr5 stones. Within the same carver the trend goes the same way but is not significant (p = 0.084). The finding can therefore be read as a difference between carving traditions rather than a gradual change in everyone; that fits a written norm, but the result rests on few workshops.
3. **The dating.** The style groups are a relative chronology with uncertain years. RAK is a broad, early group whose stones stand mostly in Götaland; the model holds location constant, and Uppland on its own shows the same direction, but there RAK is largely Gunnar. With drawn years the change becomes 22–33 percentage points instead of 48.
4. **The measurement.** The trait rests on Rundata's normalisation and transliteration, word by word, without a reading of our own. The coding check changes nothing, but signed stones alone give no significant result (106 stones).
5. **What the spelling means.** Whether *stin* denotes a monophthong or is a shortened way of writing the diphthong is the very question the result touches but cannot decide.
6. **Literature we have not read.** Källström (2010) on Åsmund, Öpir and Önjut and Åhlén's (1997) monograph on Öpir bear on reading Öpir's spelling as a norm.

**What would overturn the finding.** A dating independent of the ornament in which the late stones of the Mälaren region had a monophthong spelling as often as the early ones, or Öpir's Pr5 stones turning out to stand in a district where every carver wrote the diphthong, so that the difference is one of place and not of time.

## 7. Conclusion

The monophthong spelling becomes rarer on the runestones, from 69 to 22 percent with location and carver constant, and the change withstands every sensitivity analysis. It runs against the sound change the spelling has been used to date, and the late Upplandic spelling *stain* probably reflects a written norm. The next step is to test the change within carving schools across generations and against a dating independent of the ornament.

## Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se). Code state: commit 4f80a63 in https://github.com/Precatio/runes (Vitki 3.0.0 plus later changes); the check of the halves and the report's builder for the versions of 2026-11-05 from commit 6eae0af. The code and result files are there under GPL-3.0.

- The trait: `src/language_profile.py`; the change, the halves and the checks of the measurement: `r/change.R`; the sensitivity analyses: `r/robustness.R` and `src/robustness.py` (`METHODS.md`, 1v); the results under the material's version `e83202e01a5e6d27`.
- The coding check: `utdata/monoftong/kodning.py` and `kodning.R`. The deviation of the halves: `utdata/monoftong/halvor.R` and `halvor.json` (added 2026-11-05 in commit 6eae0af, together with the builder of version 2).
- The report is built by `utdata/blogg/build/build_rapport_monoftong.py`, which stops if any figure disagrees. The template: `utdata/rapporter/MALL.md`, version 4.
- In Vitki: Statistics, Change. Glossary: https://vitkirunor.se/en/forskningsrapporter/begrepp.

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## Change log

- **Version 2, 2026-11-05.** Corrections before archiving in Zenodo: the abstract also gives the smaller decline when the years are drawn within the style groups' intervals (22–33 percentage points); the sentence on the halves in the sensitivity analyses was wrong (they are computed by another part of the program, and nine stones, not three, fall in another half); the difference in percentage points is now given in both languages; Källström (2010) and Åhlén (1997) are named as unread. Added code state, data versions, the template's version (4), the glossary and a source in the chart; the text is shortened. No analysis was redone and no earlier figure changed.
- **Version 3, 2026-11-05.** Corrections after a second review before archiving in Zenodo. Weakness 4 again says that the trait rests on Rundata's normalisation and transliteration, word by word, without a reading of our own; version 2 removed this without saying so. The check files of the halves (halvor.R and halvor.json) are not in the stated code state but in commit 6eae0af, together with the report's builder, and Data and code now says so. Version 2 also removed, without saying so, that the share hardly moves between about 1030 and 1070 (from the abstract; it remains in 5.1 and weakness 1), that the words spelled with a alone or without a vowel show no monophthong, that the change is not confined to the word for stone (5.4; the figures remain) and that the e-spelling was not available everywhere from the start because dotted runes become more common during the period (5.5). No analysis was redone and no earlier figure changed.

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**Cite as**

Kvant, V. (2026). Monophthong spellings become rarer on the runestones – from 69 to 22 percent between about 1000 and 1115. Research report, report version 3; analysis in Vitki 3.0.0. https://vitkirunor.se/en/forskningsrapporter/monoftongstavningen-blir-ovanligare